

DHRUV RATAN JAYASWAL

BACKEND ENGINEER

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SUMMARY

Backend Engineer with expertise in high-performance APIs and data pipeline architecture. Experienced building production systems with TypeScript, PostgreSQL, and event-driven architectures. Focused on API design, database optimization, and infrastructure reliability

TECHNICAL SKILLS

TypeScript, Express, FastAPI, MongoDB, JWT, Mongoose, Drizzle, Postman, PostgreSQL, Redis, Pooling, Nginx, Docker, AWS EC2 and S3, Node, Elysia, Uvicorn, Git, React, Next, Tailwind, Zod

PROFESSIONAL EXPERIENCE

Codal Virtualwave Pvt Ltd from **April 2025** to **July 2025**

- Architected RESTful APIs using Elysia and Bun, handling payment gateway integration with Cashfree, implemented JWT-based authentication with role-based access control (RBAC)
- Optimized PostgreSQL queries and designed database schema with Drizzle ORM, achieved 40% reduction in query latency through caching using Redis, indexing and pagination.
- Containerized microservices with Docker and set up CI/CD pipeline for automated deployments.
- Managed AWS S3 bucket policies and lifecycle management for scalable file storage

PROJECTS

1. **Alluvium** is an AI powered video & document analyzer <https://el-alluvium.vercel.app>
 - Architected FastAPI backend with semantic search using ChromaDB and Redis caching, achieved 300-500ms p95 latency on 100K+ vectors, with RBAC access for user/viewer/admin.
 - Designed async background worker system with long polling for document processing, handles 500MB+ files without blocking requests.
 - Implemented Redis dual-layer caching for auth tokens (40% overhead reduction) and query results (65-70% hit ratio).
 - Set up Nginx load balancing on AWS EC2, deployed across multiple FastAPI instances.
 - Optimized AWS S3 documents with versioning, encryption, and efficient retrieval strategies.
2. **Xegality** is an Attorney/Lawyer Management Platform <https://xegality.com>
 - Built Bun(Node) + Elysia(Express) backend for attorney-client case management with <50ms query latency through PostgreSQL optimization and strategic indexing, pagination, caching.
 - Integrated Razorpay payment processing with idempotent transaction handling and webhook reconciliation for attorney billing.
 - Implemented Redis caching for auth and case/document data, with AWS S3 for media files.
 - Integrated AI models for case analysis, persisted model responses and derived knowledge in database for incremental learning.
 - Designed five-tier RBAC (attorney, paralegal, litigant, judge, admin).

EDUCATION

B.Tech in Computer Science Engineering from **Poornima University** with **8.45 CGPA** **2022 - 2026**
Senior and Senior Secondary from **DAV Centenary Public School** with **85%** **2018 - 2021**

CERTIFICATIONS

<u>FullStack - META</u>	<u>Javascript - Hackerrank</u>	<u>DSA (Java) - Scaler</u>	<u>Aggregation - Mongo</u>
<u>Node - Simplilearn</u>	<u>React - Scaler</u>	<u>Backend - Celebal</u>	<u>Document Model - Mongo</u>